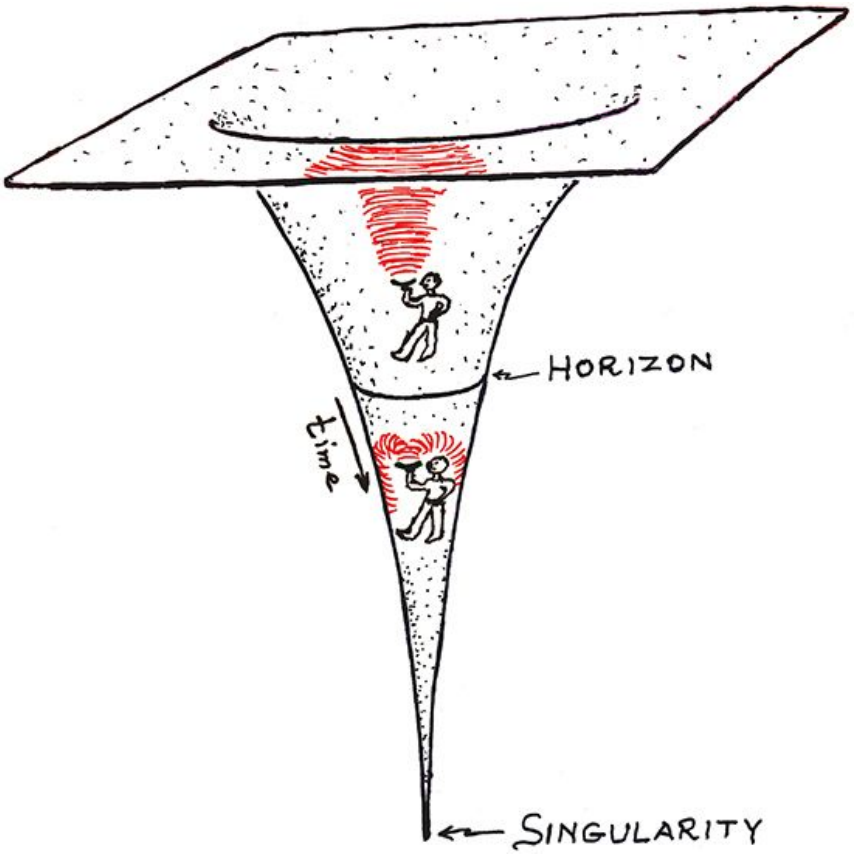


Fig. 5.3. Signals I send after crossing the event horizon can't get out. *Note:* Because one space dimension is removed from this diagram, I am a two-dimensional Kip, sliding down the warped two-dimensional surface, part of our brane. *[My own hand sketch.]*

If I fall into a black hole carrying a microwave transmitter, then once I pass through the hole's *event horizon*, I'm pulled inexorably on downward, into the hole's singularity. And any signals I try to transmit in any manner whatsoever get pulled down with me. Nobody above the horizon can ever see the signals I send after I cross the horizon. My signals and I are trapped inside the black hole. (See Chapter 28 for how this plays out in *Interstellar*.)

This trapping is actually caused by the hole's time warp. If I hover above the black hole, supporting myself by the blast of a rocket engine, then the closer I am to the horizon, the more slowly my time flows. At the horizon itself, time slows to a halt and, therefore, according to